CURRICULUM VITAE

Dr. Junke Wang

State Key Laboratory of Luminescent Materials and Devices, School of Materials Science and Engineering
South China University of Technology
Guangzhou, China

ORCID: 0000-0002-5849-7297
Wjk090426@scut.edu.cn
+44 7923973773

EDUCATION

PhD in Chemistry – Eindhoven University of Technology, The Netherlands **2016 – 2020**

Dissertation: "Interfacial, Compositional and Morphological Engineering for Single- and Multi-junction Perovskite Solar Cells"

Supervisor: Prof. René A.J. Janssen

MSc in Chemical Engineering – Eindhoven University of Technology, The Netherlands **2014 – 2016**

Dissertation: "Sequential Deposition Method for Planar Perovskite Solar Cells"

With great appreciation

BE in Materials Science and Engineering – Central South University, China **2009 – 2013**

Cum Laude

ACADEMIC APPOINTMENTS

Professor, South China University of Technology, Guangzhou, China **2026 – Present**

- Principle investigator of a research team working on next-generation photovoltaics and optoelectronics

Postdoc, Marie-Curie Fellow, University of Oxford (Supervisor: Prof. Henry J. Snaith) **2023 – 2026**

- Lead research on hybrid wide-bandgap semiconductors for high-efficiency multi-junction solar cells
- Secured £200,512 funding from the UKRI for Guarantee Funding for Marie-Curie Fellowship

Postdoc, jointly between University of Toronto (Prof. Edward H. Sargent) and Eindhoven University of Technology (Prof. René A.J. Janssen) **2020 – 2023**

- Developed stable high-bandgap perovskite materials for all-perovskite triple-junction solar cells
- Conducted research on perovskite-based tandem photovoltaics for solar-driven water splitting

PhD Candidate, Eindhoven University of Technology (Supervisor: Prof. René A.J. Janssen) **2016 – 2020**

- Conducted research on hybrid semiconductors for solar energy storage, photodiodes, and light-driven CO_2 reduction
- Developed prototype all-perovskite triple-junction solar cells

- Supervised PhD/Master students and contributed to teaching undergraduate chemistry courses

RESEARCH INTERESTS

- Perovskite multijunction solar cells
- Light-trapping nanostructures for photovoltaics
- Operando characterization techniques for optoelectronics (XRD, XPS, TEM, CL, optical Spectroscopy)
- Sustainable materials and renewable energy technologies

AWARDS & GRANTS

Marie Curie Postdoctoral Fellowship (UKRI Guarantee, EP/Y029216/1, £200,512)	**2023**
The 2020 Chinese Government Award for Outstanding Self-financed Students Abroad (US$6k)	**2021**
Amandus H. Lundqvist Scholarship Program, Eindhoven University of Technology, The Netherlands (~€36k)	**2014**

TEACHING EXPERIENCE

- Teaching assistant for third-year undergraduates in chemistry (Design-based Energy Course, 3 semesters, TU/e)
- Co-supervised 2 masters' thesis and 2 bachelors' thesis (TU/e)
  Alessandro Caiazzo, dissertation "*Multidimensional perovskite solar cells*"
  Hiten Tarpara, dissertation "*Pb-Sn based ideal-bandgap perovskite solar cell*"
  Ronald de Bruijne, dissertation "*Fabrication of stable pure-phase $FAPbI_3$ perovskites for photovoltaic applications*"
  Jasper Slingerland, dissertation "*Development of All-Bromide Pb-Sn Hybrid Wide Bandgap Perovskite Solar Cells*"

INTERNATIONAL CONFERENCES

Invited talk at the MATSUS Fall Conference 2026, Spain	**2026**
Invited talk at the Solar Energy Seminar Series at LMU Munich, Germany	**2026**
Oral presentation at the MRS Fall Meeting 2025, United States	**2025**
Oral presentation at the TandemPV Workshop 2025, Belgium	**2025**
Invited talk at the Solar Energy Seminar at Helmholtz-Zentrum Berlin, Germany	**2025**
Oral presentation at the MRS Fall Meeting 2024, United States	**2024**
Oral presentation at the SPIE Optics + Photonics 2024, United States	**2024**
Oral presentation at the TandemPV Workshop 2024, The Netherlands	**2024**
Oral presentation at the 6th International Conference on Perovskite Solar Cells and Optoelectronics, United Kingdom	**2023**
Oral presentation at the MRS Fall Meeting 2022, Virtual	**2022**
Oral presentation at the MRS Spring Meeting 2022, Virtual	**2022**
Oral presentation at the nanoGe Spring Meeting, Virtual	**2021**

Oral presentation at the 5th International Conference on Perovskite Solar Cells and Optoelectronics, Switzerland **2019**

Poster award at the International Conference on Hybrid and Organic Photovoltaics, Spain **2018**

MEDIA OUTREACH

CNN.com	Razor-thin solar panels could be 'ink-jetted' onto your backpack or phone for cheap clean energy
MSN.com	Solar energy breakthrough could mean solar panels will be a thing of the past
PV Magazine	All-perovskite triple-junction solar cell achieves 25.1% efficiency via new passivation technique
PV Magazine	Triple junction perovskite cell hits 16.8% efficiency

PUBLICATION LIST

- **Wang J.**,† Hu S.,† Gu X.,† Truong M.A.,† Yang Y.,† Ning Z., Wakamiya A., Snaith H.J., Chen H. et al. *Homogenised Optoelectronic Properties in Perovskites: Achieving High-Efficiency Solar Cells with Common Chloride Additives*. Journal of the American Chemical Society, 2026, 148, 6229. https://pubs.acs.org/doi/10.1021/jacs.5c18303
- Liu Y.,† **Wang J.**,† Snaith H.J., Chen S. et al. *Stabilized perovskite ink for scalable coating enables high-efficiency perovskite modules*. Science Advances, 2026, 12, eaec0915. https://www.science.org/doi/10.1126/sciadv.aec0915
- **Wang J.**,† Hu S.,† Chen Z.†, Snaith H.J. et al. *Exposing binding-favourable facets of perovskites for tandem solar cells*. Energy & Environmental Science, 2025, 18, 7680. https://pubs.rsc.org/en/content/articlelanding/2025/ee/d5ee02462e
- Hu S.,† **Wang J.**,† Wakamiya A., Snaith H.J. et al. *Steering tin–lead perovskite precursor solutions for multijunction photovoltaics.* Nature, 2025, 639, 93. https://doi.org/10.1038/s41586-024-08546-y
- **Wang J.**,† Branco B.,† Janssen R.A.J. et al. *All-perovskite tandem photovoltaics in light-driven electrochemical water splitting*. Nature Communications, 2025, 16, 174. https://doi.org/10.1038/s41467-024-55654-4
- **Wang J.**,† Zeng L.,† Zhang D.,† Maxwell A.,† Chen H.,† Janssen R.A.J., Sargent E.H. et al. *Halide homogenization for low energy loss in 2-eV-bandgap perovskites and increased efficiency in all-perovskite triple-junction solar cells*. Nature Energy, 2024, 9, 70. https://www.nature.com/articles/s41560-023-01406-5
- Datta K.,† **Wang J.**,† Zhang D., Zardetto V., Remmerswaal W.H.M., Weijtens C.H.L., Wienk M.M., Janssen R.A.J. *Monolithic All-Perovskite Tandem Solar Cells with Minimized Optical and Energetic Losses*. Advanced Materials, 2021, 2110053. https://onlinelibrary.wiley.com/doi/full/10.1002/adma.202110053
- Gómez P.,† **Wang J.**,† Más-Montoya M., Bautista D., Weijtens C.H.L., Curiel D., Janssen R.A.J. *Pyrene-based small molecular hole transport layers for efficient and stable narrow-bandgap perovskite solar cells*. Solar RRL, 2021, 5, 2100454. https://onlinelibrary.wiley.com/doi/full/10.1002/solr.202100454
- **Wang J.,** Zardetto V., Datta K., Zhang D., Wienk M.M., Janssen R.A.J. *16.8% Monolithic All-perovskite Triple-junction Solar Cells via A Universal Two-step Solution Process*. Nature Communications, 2020, 11, 5254. https://www.nature.com/articles/s41467-020-19062-8

- **Wang J.**, Datta K., Li J., Verheijen M.A., Zhang D., Wienk M.M., Janssen R.A.J. *Understanding the Film Formation Kinetics of Sequential Deposited Narrow-Bandgap Pb–Sn Hybrid Perovskite Films.* Advanced Energy Materials, 2020, 10, 2000566. https://onlinelibrary.wiley.com/doi/full/10.1002/aenm.202000566
- **Wang J.**, Datta K., Weijtens C.H.L., Wienk M.M., Janssen R.A.J. *Insights into fullerene passivation of SnO_2 electron transport layers in perovskite solar cells.* Advanced Functional Materials, 2019, 29, 1905883. https://onlinelibrary.wiley.com/doi/full/10.1002/adfm.201905883
- **Wang J.** et al. *Highly Efficient Perovskite Solar Cells Using Non-Toxic Industry Compatible Solvent System.* Solar RRL, 2017, 1, 1700091. https://onlinelibrary.wiley.com/doi/full/10.1002/solr.201700091
- Shen X., Hui W.T., Hu S., Yang F., **Wang J.**, Yao J., Louwen A., Tam B.S.T., Rong L., McMeekin D.P., Lohmann K., Yuan Q., Naylor M.C., Kober-Czerny M., Seo S., Holzhey P., Zaininger K.-A., Christoforo M.G., Carroy P., Barth V., Yeung F.S.Y., Noel N.K., Johnston M., Lin Y.-H., Snaith H.J. *Crystal-facet-directed all-vacuum-deposited perovskite solar cells*. Nature Materials, 2026, 25, 999.
- Schipper N.R.M., Aalbers G.J.W., Bellini L., Quiroz Monnens S.V., Kessels L.M., **Wang J.**, Wienk M.M., Janssen R.A.J. *The Importance of Conserving the Stoichiometry of Wide-Bandgap Perovskites in Additive Engineering*. ACS Applied Energy Materials, 2025, 8, 14486. https://pubs.acs.org/doi/full/10.1021/acsaem.5c02216
- Wang J., Hu S., Zhu H., Liu S., Zhang Z., Chen R., **Wang J.**, Shi. C., Zhang J., Liu W., Lei X., Liu B., Pan Y., Ren F., Raza. H., Zhou Q., Li S., Qiu L., Zheng G., Qin X., Zhao Z., Yang S., Li N., Li J., Wakamiya A., Liu Z., Snaith H.J., Chen W. *Mercapto-functionalized scaffold improves perovskite buried interfaces for tandem photovoltaics*. Nature Communications, 2025, 16, 4917. https://www.nature.com/articles/s41467-025-59891-z
- Nambiar R.A., McMeekin D.P., Kober-Czenry M., Smith J.A., Taddei M., Caprioglio P., Kumar A., Putland B.W., **Wang J.**, Elmestekawy K.A., Dasgupta A., Seo S., Christoforo M.G., Yao J., Graham D.J., Herz L.M., Ginger D., Snaith H.J. *Interdiffusion control in sequentially evaporated organic–inorganic perovskite solar cells*. EES Solar, 2025. https://pubs.rsc.org/en/content/articlehtml/2025/el/d5el00017c
- Wu L., Hu S., Yang F., Li G., **Wang J.**, Zuo W., Jerónimo-Rendon J.J., Turren-Cruz S.-H., Saba M., Saliba M., Nazeeruddin M.K., Pascual J., Li M., Abate A. *Resilience pathways for halide perovskite photovoltaics under temperature cycling*. Nature Review Materials, 2025. https://www.nature.com/articles/s41578-025-00781-7
- Hu S., **Wang J.**, Seo S., Snaith H.J., *V. On the stability of perovskite-containing multijunction photovoltaics*. Materials Today Electronics, 2025, 11, 100138. https://www.sciencedirect.com/science/article/pii/S277294942500004X
- Li C., Chen L., Jiang F., Song Z., Wang X., Balvanz A., Ugur E., Liu Y., Liu C., Maxwell A., Chen H., Liu Y., Wang Z., Xia P., Li Y., Fu S., Sun N., Grice C.R., Wu X., Fink Z., Hu Q., Zeng L., Jung E., **Wang J.**, Park S.M., Luo D., Chen C., Shen J., Han Y., Perini C.A.R., Correa-Baena J.-P., Lu Z.-H., Russell T.P., De Wolf S., Kanatzidis M.G., Ginger D.S., Chen B., Yan Y., Sargent E.H. *Diamine chelates for increased stability in mixed Sn–Pb and all-perovskite tandem solar cells*. Nature Energy, 2024, 9, 1388. https://doi.org/10.1038/s41560-024-01613-8

- Chen H., Liu C., Xu J., Maxwell A., Zhou W., Yang Y., Zhou Q., Bati A.S.R., Wan H., Wang Z., Zeng L., **Wang J.**, Serles P., Liu Y., Teale S., Liu Y., Saidaminov M.I., Li M., Rolston N., Hoogland S., Filleter T., Kanatzidis M.G., Chen B., Ning Z., Sargent E.H. *Improved charge extraction in inverted perovskite solar cells with dual-site-binding ligands*, Science, 2024, 384, 189. https://www.science.org/doi/full/10.1126/science.adm9474
- Maxwell A., Chen H., Grater L., Li C., Teale, S., **Wang J.**, Zeng L., Wang Z., Park S.M., Vafaie M., Sidhik S., W Metcalf I.W., Liu L., Mohite A.D., Chen C., Sargent E.H. *All-Perovskite Tandems Enabled by Surface Anchoring of Long-Chain Amphiphilic Ligands*, ACS Energy Letters, 2024, 9, 520. https://pubs.acs.org/doi/full/10.1021/acsenergylett.3c02470
- Wang Z., Zeng L., Zhu T., Chen H., Chen B., Kubicki D.J., Balvanz A., Li C., Maxwell A., Ugur E., Reis, R.D., Cheng M., Yang G., Subedi B., Luo D., Hu J., **Wang J.**, Teale S., Mahesh S., Wang S., Hu S., Jung E.D., Wei M., Park S.M., Grater L., Aydin E., Song Z., Podraza N.J., Lu Z.-H., Huang J., Dravid V.P., De Wolf S., Yan Y., Grätzel M., Kanatzidis M.G., Sargent E.H. *Suppressed phase segregation for triple-junction perovskite solar cells*, Nature, 618, 74. https://www.nature.com/articles/s41586-023-06006-7
- Zhao Y., Datta K., Phung N., Bracesco A.E.A., Zardetto V., Paggiaro G., Liu H., Fardousi M., Santbergen R., Moya P.P., Han C., Yang G., **Wang J.**, Zhang D., van Gorkom B.T., van der Pol T.P.A., Verhage M., Wienk M.M., Kessels W.M.M., Weeber A., Zeman M., Mazzarella L., Creatore M., Janssen R.A.J., Isabella O. *Optical simulation-aided design and engineering of monolithic perovskite/silicon tandem solar cells*, ACS Applied Energy Materials, 2023, 6, 5217. https://pubs.acs.org/doi/full/10.1021/acsaem.3c00136
- Caiazzo A., Maufort A., van Gorkom B.T., Remmerswaal W.H.M., Orri J.F., Li J., **Wang J.**, van Gompel W.T.M., van Hecke K., Kusch G., Oliver R.A., Ducati C., Lutsen L., Wienk M.M., Stranks S.D., Vanderzande D., Janssen A.J., *3D Perovskite Passivation with a Benzotriazole-Based 2D Interlayer for High-Efficiency Solar Cells*, ACS Applied Energy Materials, 2023, 6, 3933. https://pubs.acs.org/doi/full/10.1021/acsaem.3c00101
- Chen H., Maxwell A., Li C., Teale S., Chen B., Zhu T., Ugur E., Harrison G., Grater L., **Wang J.**, Wang Z., Zeng L., Park S.M., Chen L., Serles P., Awni R.A., Subedi B., Zheng X., Xiao C., Podraza N.J., Filleter T., Liu C., Yang Y., Luther J.M., De Wolf S., Kanatzidis M.G., Yan Y., Sargent E.H. *Regulating surface potential maximizes voltage in all-perovskite tandems*, Nature 2023, 613, 676. https://www.nature.com/articles/s41586-022-05541-z
- Más-Montoya M., Gómez P., **Wang J.**, Janssen R.A.J., Curiel D. *Small molecule dopant-free dual hole transporting material for conventional and inverted perovskite solar cells*, Materials Chemistry Frontiers, 2023, 7, 4019. https://pubs.rsc.org/en/content/articlehtml/2023/qm/d3qm00425b
- Tulus, **Wang J.**, Galagan Y., von Hauff E. *Quantifying electrochemical losses in perovskite solar cells*, Journal of Materials Chemistry C, 2023, 11, 2911. https://pubs.rsc.org/en/content/articlehtml/2023/tc/d2tc03486g
- Bin H., Datta K., **Wang J.**, van der Pol T.P.A., Li J., Wienk M.M., Janssen R.A.J., *Finetuning Hole-Extracting Monolayers for Efficient Organic Solar Cells*. ACS Applied Materials & Interfaces, 2022, 14, 16497. https://pubs.acs.org/doi/full/10.1021/acsami.2c01900
- Ollearo R., **Wang J.**, Dyson M.J., Weijtens C.H.L., Fattori M., van Gorkom B.T., van Breemen A.J.J.M., Meskers S.C.J., Janssen R.A.J., Gelinck G.H. *Ultralow dark current in near-infrared perovskite photodiodes enabled*

by reducing charge injection and interfacial charge generation. Nature Communications, 2021, 12, 7277. https://www.nature.com/articles/s41467-021-27565-1

- Bin H., **Wang J.**, Li J., Wienk M.M., Janssen R.A.J. *Efficient Electron Transport Layer Free Small-Molecule Organic Solar Cells with Superior Device Stability*. Advanced Materials, 2021, 33, 2008429. https://onlinelibrary.wiley.com/doi/full/10.1002/adma.202008429
- Más-Montoya M., Curiel D., **Wang J.**, Bruijnaers B.J., Janssen R.A.J., *Use of Sodium Diethyldithiocarbamate to Enhance the Open-Circuit Voltage of $CH_3NH_3PbI_3$ Perovskite Solar Cells*. Solar RRL, 2021, 5, 2000811. https://onlinelibrary.wiley.com/doi/full/10.1002/solr.202000811
- Más-Montoya M., Gómez P., Curiel D., Da Silva I., **Wang J.**, Janssen R.A.J. *A Self-Assembled Small Molecule-Based Hole Transporting Material for Inverted Perovskite Solar Cells*. Chemistry – A European Journal, 2020, 26, 10276. https://chemistry-europe.onlinelibrary.wiley.com/doi/full/10.1002/chem.202000005
- Esiner S., **Wang J.**, Janssen R.A.J. *Light-Driven Electrochemical Carbon Dioxide Reduction to Carbon Monoxide and Methane Using Perovskite Photovoltaics*. Cell Reports Physical Science, 2020, 1, 100058. https://www.cell.com/cell-reports-physical-science/fulltext/S2666-3864(20)30052-7
- Wang Q., Li M., Zhang X., Qin Y., **Wang J.**, Zhang J., Hou J., Janssen R.A.J., Geng Y. *Carboxylate-Substituted Polythiophenes for Efficient Fullerene-Free Polymer Solar Cells: The Effect of Chlorination on Their Properties*. Macromolecules, 2019, 52, 4464. https://pubs.acs.org/doi/full/10.1021/acs.macromol.9b00793
- Muscarella L.A., Petrova D., Cervasio R.J., Farawar A., Lugier O., McLure C., Slaman M.J., **Wang J.**, Ehrler B., Von Hauff E., Williams R.M. *Air-stable and oriented mixed lead halide perovskite (FA/MA) by the one-step deposition method using zinc iodide and an alkylammonium additive*, ACS Applied Materials & Interfaces, 2019, 11, 17555. https://pubs.acs.org/doi/full/10.1021/acsami.9b03810
- Li M., Li J., Di Carlo Rasi D., Colberts F.J.M, **Wang J.**, Heintges G.H.L., Wienk M.M., Janssen R.A.J. *The Impact of Device Polarity on the Performance of Polymer–Fullerene Solar Cells*. Advanced Energy Materials, 2018, 8, 1800550. https://onlinelibrary.wiley.com/doi/full/10.1002/aenm.201800550
- Xu X., Li Z., **Wang J.**, Lin B., Ma W., Xia Y., Andersson M.R., Janssen R.A.J., Wang. E. *High-performance all-polymer solar cells based on fluorinated naphthalene diimide acceptor polymers with fine-tuned crystallinity and enhanced dielectric constants*. Nano Energy, 2018, 45, 368. https://www.sciencedirect.com/science/article/pii/S2211285518300132
- **Wang J.**, Zhang H., Ma Z., Zhang Y., Li Z. *Abnormal resistivity-temperature characteristic in fluorite type Bi/K-substituted ceria ceramics*, 2016, 27, 6419. https://link.springer.com/article/10.1007/s10854-016-4580-8